

Notes: Autor, Dorn, and Hanson (AER: Insights, 2019)

When Work Disappears: Manufacturing Decline and the Falling Marriage Market Value of Young Men

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1 Big picture

- **Question.** How do large local labor-demand shocks that disproportionately hit young men change marriage, fertility, and children's living circumstances?
- **Setting.** The paper studies the rise in Chinese import competition across 722 U.S. commuting zones (CZs), using stacked decade-equivalent changes for 1990–2000 and 2000–2014.
- **Core challenge.** Areas hit by manufacturing decline also differ in many other ways. A simple correlation between local job loss and family outcomes could therefore reflect local demand conditions, migration, changing demographics, or broader regional decline rather than a causal effect of men's labor-market deterioration.
- **Main conceptual contribution.** The paper exploits *gender-specific* components of local trade shocks. Because some manufacturing industries are more male intensive and others more female intensive, the same China shock can be decomposed into a shock that mainly hits men and one that mainly hits women.
- **Main mechanism.** In Becker's household-specialization logic, marriage and fertility are more attractive when men's relative earnings capacity is high. In Wilson's broader sociological account, an absolute decline in the pool of economically secure young men weakens marriageability and family stability.
- **Main empirical design.** Regress changes in male and female employment, earnings, idleness, mortality, marriage, fertility, and child outcomes on CZ-level China import exposure, instrumented with the growth of Chinese exports to other developed countries.
- **Main baseline result.** Trade shocks reduce male relative to female employment and earnings. A one-unit trade shock lowers the male-female employment gap by 0.65 percentage points and compresses the male-female earnings gap by \$672 at the 25th percentile, \$445 at the median, and \$847 at the 75th percentile.
- **Main social-function result.** Trade shocks raise male idleness and premature male mortality. Among young adults, a one-unit trade shock raises excess male mortality by 64.4 deaths per 100,000 per decade.
- **Main family result.** The same shocks reduce marriage and fertility and worsen children's living circumstances. For women ages 18–39, a one-unit trade shock lowers the currently married share by 0.95 percentage points and raises the never-married share by 1.16 points. It also raises child poverty by 0.61 points.
- **Key heterogeneity result.** Shocks centered on male-intensive industries produce the adverse effects. Shocks centered on female-intensive industries often have the opposite sign.
- **Main credibility result.** The paper uses an established China-trade instrument, decomposes shocks by gender intensity, includes extensive CZ controls, and reports falsification evidence that the shock does not predict employment in the 1980s.
- **Economic takeaway.** The decline of blue-collar male employment changes not just labor-market outcomes but family structure itself. Trade shocks that reduce men's economic standing

lower marriageability, reduce fertility, and increase the share of children raised in poor single-parent households.

2 Data and measurement (institutional context and key objects)

2.1 Why this setting is useful

- The paper studies a nationally consistent local-labor-market shock with substantial cross-CZ variation.
- The mechanism of interest is not just whether manufacturing contracts, but whether male earnings capacity falls *relative* to female earnings capacity.
- The China shock is especially useful because it is concentrated in manufacturing, varies by local industry mix, and can be decomposed by gender intensity.

Interpretation. The setting gives the paper a way to connect a plausibly exogenous labor-demand shock to gender-specific economic status and then to family outcomes.

2.2 Data sources and sample construction

- Local labor markets are measured at the level of 722 mainland U.S. commuting zones.
- Trade exposure uses UN Comtrade imports and U.S. industry data harmonized to four-digit SIC industries.
- Industry employment shares come from County Business Patterns.
- Gender shares within industries and CZs come from census IPUMS data.
- Most household outcomes use census IPUMS data for 1990 and 2000, plus pooled ACS data for 2013–2015.
- Mortality outcomes use Vital Statistics death-certificate data for 1990–2015.
- Fertility outcomes extend through 2016.

Interpretation. The paper combines detailed administrative and census-style sources so that the same trade shock can be traced through labor-market, demographic, and family outcomes.

2.3 Overall CZ trade exposure

The core China-import-exposure measure for CZ i over period τ is

$$\Delta IP_{i\tau}^{cu} = \sum_j \frac{L_{ij,90}}{L_{i,90}} \Delta IP_{j\tau}^{cu}, \quad (1)$$

where $L_{ij,90}/L_{i,90}$ is industry j 's share in CZ employment in 1990 and $\Delta IP_{j\tau}^{cu}$ is the change in Chinese import penetration in U.S. industry j over the period.

Import penetration in industry j is defined as

$$\Delta IP_{j\tau}^{cu} = \frac{\Delta M_{j\tau}^{cu}}{Y_{j,91} + M_{j,91} - X_{j,91}}, \quad (2)$$

that is, the growth in U.S. imports from China divided by initial absorption.

Interpretation. A CZ is more exposed if it was initially specialized in industries where Chinese import penetration later rose strongly.

2.4 Gender-specific trade shocks

The paper splits the overall trade shock into male- and female-intensive components:

$$\Delta IP_{i\tau}^{m,cu} = \sum_j (1 - f_{ij,90}) \frac{L_{ij,90}}{L_{i,90}} \Delta IP_{j\tau}^{cu}, \quad (3)$$

$$\Delta IP_{i\tau}^{f,cu} = \sum_j f_{ij,90} \frac{L_{ij,90}}{L_{i,90}} \Delta IP_{j\tau}^{cu}, \quad (4)$$

where $f_{ij,90}$ is the female employment share in industry j and CZ i .

Interpretation. The decomposition asks whether the shock primarily loads on male jobs or female jobs. That is what lets the paper test Becker-style relative-earnings predictions directly.

2.5 Instrument for the trade shock

The paper instruments for U.S. China import exposure using the contemporaneous growth of Chinese exports to eight other developed countries:

$$\Delta IP_{i\tau}^{co} = \sum_j \frac{L_{ij,80}}{L_{i,80}} \Delta IP_{j\tau}^{co}. \quad (5)$$

The eight comparison countries are Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland.

Interpretation. The instrument isolates the common component of Chinese export growth driven by China-side productivity and trade-cost changes rather than by U.S.-specific demand shifts.

2.6 Main outcome variables

The paper studies several outcome blocks.

Labor-market outcomes:

- Manufacturing employment as a share of population, ages 18–39.

- Male-female differential in employment, unemployment, and nonparticipation.
- Male-female differential in annual earnings at the 25th percentile, median, and 75th percentile.
- Male-female differential in idleness among ages 18–25, where idleness means not employed and not in school.

Availability and distress of young men:

- Male share of the resident population ages 18–39 and 18–25.
- Male-female mortality differential ages 20–39, total and by cause.

Family outcomes:

- Women’s marital status: currently married, previously married, never married.
- Births per 1,000 women ages 20–39.
- Fraction of women with children in the household.
- Fraction of mothers who are unmarried.
- Fraction of children living below the poverty line.
- Women’s and children’s household structure.

Interpretation. The paper moves in a deliberate sequence: first men’s relative earnings capacity, then men’s marriage-market value, then family formation, then children’s circumstances.

2.7 Controls, weighting, and estimation sample

- Observations are stacked over two periods: 1990–2000 and 2000–2014, giving 1,444 CZ-period observations in the main tables.
- Regressions include a dummy for the 2000–2014 period.
- Controls include census-division trends, the initial manufacturing share, routine-occupation and offshorability indices, race shares, education, foreign-born share, and the fraction of women employed.
- Regressions are weighted by period length times each CZ’s share of the start-of-period mainland U.S. population.
- Standard errors are clustered by state.

Interpretation. The paper does not rely on a raw cross section. It uses weighted stacked first differences with rich controls and state-clustered inference.

2.8 Descriptive facts

Some baseline facts are useful for interpretation.

- In 1990, 17.4% of young men ages 18–39 and 8.7% of young women ages 18–39 worked in manufacturing.
- The mean male-female employment gap in 1990 was 14.64 percentage points.
- The male-female annual earnings gap in 1990 was \$6,926 at the 25th percentile, \$13,376 at the median, and \$17,489 at the 75th percentile.
- The cumulative male-female mortality gap ages 20–39 over 1990–2015 averaged 936 deaths per 100,000 per decade.
- In 1990, 53.05% of women ages 18–39 were currently married and 34.84% were never married.
- In 1990, 17.99% of children lived in poverty, with huge differences by household type: 8.7% in married households, 42.3% in cohabiting households, and 47.4% in single-headed households.

Interpretation. These starting values help show why even seemingly small coefficient estimates can matter economically.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework and hypotheses

The paper combines two ideas.

- **Becker (relative economic stature).** If male earnings fall relative to female earnings, the gains from household specialization decline. Marriage and fertility should fall.
- **Wilson (absolute male economic decline).** If the pool of economically secure young men shrinks, male idleness, distress, and family instability should rise.

These mechanisms generate three empirical predictions:

- shocks that reduce male relative employment and earnings should reduce marriage and fertility;
- they should worsen men’s social function and marriageability through idleness and mortality;
- they should increase the share of mothers who are unmarried and children living in poor single-parent households.

Interpretation. The gender-specific decomposition matters because Becker predicts opposite family effects from shocks to male versus female earnings capacity.

3.2 Baseline estimating equation

For outcome Y_{sit} in gender group s , the paper estimates

$$\Delta Y_{sit} = \alpha_t + \beta_1 \Delta IP_{it}^{cu} + X'_{it} \beta_2 + e_{sit}. \quad (6)$$

For gender-specific regressions, the overall trade shock is replaced by the male and female components, $\Delta IP_{i\tau}^{m,cu}$ and $\Delta IP_{i\tau}^{f,cu}$.

Interpretation. The baseline coefficient tells how a decade-equivalent change in local China exposure changes local outcomes. The gender-specific version asks whether the effect runs through the side of the labor market hit more strongly.

3.3 Why stacked first differences are used

The panel stacks 1990–2000 and 2000–2014 changes and includes a period dummy.

- This removes time-invariant CZ factors.
- It focuses on medium-run local changes rather than short-run cyclical noise.
- It aligns the labor-market, mortality, and household outcomes to the same broad trade-shock periods.

Interpretation. The paper is about medium-run social consequences of structural labor-demand shocks, not month-to-month responses.

3.4 Outcome-specific specifications

The same basic design is applied across different outcomes.

- **Manufacturing employment:** estimated separately for all young adults, young men, and young women.
- **Relative employment status:** the dependent variable is the male-female difference in employment, unemployment, or nonparticipation.
- **Relative earnings:** estimated at unconditional percentiles using the Chetverikov-Larsen-Palmer group-treatment IV quantile approach.
- **Idleness:** measured for ages 18–25 across mutually exclusive statuses: employed, not employed but in school, and not employed and not in school.
- **Mortality:** dependent variables are cumulative male-female death-rate differentials per decade, total and by cause.
- **Marriage and fertility:** dependent variables are changes in women’s marital-status shares and births per 1,000 women.

Interpretation. The paper uses the same shock and identification logic throughout, which makes the family results read as consequences of the same initial labor-market shock.

3.5 Manufacturing employment and relative-status results

The most important estimates from Table 1 are:

- A one-unit trade shock lowers manufacturing employment as a share of population by 1.06 percentage points overall, by 0.99 points for men, and by 1.09 points for women.
- But beyond manufacturing, the same shock lowers the male-female employment gap by 0.65 points, raises the male-female unemployment gap by 0.19 points, and raises the male-female NILF gap by 0.46 points.
- The male-female earnings gap shrinks by \$672 at the 25th percentile, \$445 at the median, and \$847 at the 75th percentile.

Interpretation. Even though manufacturing employment falls for both sexes, the broader labor-market spillovers are more damaging for men. That is the paper’s measure of declining male relative economic stature.

3.6 Gender-specific shock results

When the shock is decomposed by gender intensity:

- a one-unit shock to male-intensive industries lowers male manufacturing employment by 2.59 points and has little effect on female manufacturing employment;
- a one-unit shock to female-intensive industries lowers female manufacturing employment by 2.56 points and slightly raises male manufacturing employment;
- the male-intensive shock lowers the male-female employment gap by 3.13 points, while the female-intensive shock raises it by 2.17 points.

Interpretation. This is the paper’s cleanest evidence that relative male earnings capacity, not just local manufacturing decline in general, is central to the family results.

3.7 Distributional earnings and idleness

Table 1 and Figure 1 show that the earnings effects are present throughout the male earnings distribution, but relatively strongest near the bottom.

- In Figure 1, male earnings losses are larger than female earnings losses from roughly the 15th to the 95th percentile.
- Expressed as a share of baseline male earnings, the male-female gap compresses by roughly 2 percentage points at the median, 4 points near the 35th percentile, and 6 points near the 20th percentile.
- Among ages 18–25, the entire rise in nonparticipation comes from idleness: employment falls by 0.64 points relative to women, schooling barely changes, and idleness rises by 0.66 points.

Interpretation. The shock does not mainly push young men back into school; it pushes them into inactivity.

3.8 Male availability and mortality

Table 2 studies the supply of young men in local marriage markets and male distress.

- A one-unit trade shock lowers the male share of the resident population by about 0.25 percentage points among ages 18–39 and 18–25.
- It raises the male-female mortality differential ages 20–39 by 64.4 deaths per 100,000 per decade.
- By cause, the largest increases are in drug and alcohol poisoning (19.5), HIV/AIDS (21.6), and homicide (14.0). Effects on suicide, accidents, and all other causes are small.
- Male-intensive shocks generate much larger mortality effects than female-intensive shocks.

Interpretation. The shock reduces not only men’s earnings but also their local presence and social functioning, which plausibly lowers marriage-market value further.

3.9 Marriage, fertility, and child outcomes

Table 3 provides the main family-outcome estimates.

- A one-unit trade shock lowers the share of women ages 18–39 who are currently married by 0.95 points, lowers the previously married share by 0.21 points, and raises the never-married share by 1.16 points.
- Fertility falls by 1.54 births per 1,000 women ages 20–39.
- The share of women with children in the household falls by 0.66 points, while the share of mothers who are unmarried rises by 0.52 points.
- The share of children living below the poverty line rises by 0.61 points.
- The fraction of children living in married two-parent households falls by 0.35 points, while the fraction living with a single parent and no partner rises by 0.30 points.

Interpretation. Marriage falls more than motherhood falls. That is why child poverty and single-parent living arrangements worsen.

3.10 Why male and female shocks have opposite family effects

The gender-specific estimates in Table 3 are central.

- A male-intensive shock reduces the currently married share by 3.57 points, raises never-married by 4.23 points, lowers fertility by 4.65 births per 1,000 women, raises the share of mothers who are unmarried by 3.28 points, and raises child poverty by 2.13 points.
- A female-intensive shock moves these outcomes in the opposite direction: marriage rises, fertility rises, mothers are less likely to be unmarried, and child poverty falls.

Interpretation. This sign reversal is exactly what a Becker-style relative-earnings mechanism predicts.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the estimates

Identification comes from cross-CZ differences in initial industry composition interacted with national growth in Chinese import penetration, instrumented by Chinese export growth to other developed countries.

Interpretation. Two CZs differ in exposure not because one happens to lose jobs for idiosyncratic local reasons, but because they were initially specialized in different industries that faced different China shocks.

4.2 Why the IV is plausible

The exclusion restriction is that the common component of Chinese export growth to the United States and to other developed countries is driven by China-side productivity and trade-cost changes, not by demand shocks specific to particular U.S. CZs.

Interpretation. The instrument is designed to isolate supply-driven China exposure rather than domestic demand conditions.

4.3 Why the gender decomposition helps

The paper's most convincing family evidence is not just that manufacturing decline hurts marriage. It is that:

- shocks to male-intensive employment reduce marriage and fertility and worsen child outcomes;
- shocks to female-intensive employment often increase marriage and fertility and improve child outcomes.

Interpretation. This is difficult to reconcile with a generic local-decline story. It fits much better with a mechanism running through relative male economic stature.

4.4 Controls for local confounds

The paper includes controls for:

- initial manufacturing share,
- region-specific time trends,
- demographic composition,
- routine-task and offshorable-employment exposure,

- and the fraction of women employed.

Interpretation. These controls make the comparison more demanding by netting out several competing explanations for changing employment and family structure.

4.5 Falsification and timing logic

The authors report that the China-trade shock does not negatively predict employment in the 1980s and has the opposite sign in the 1970s, when labor-intensive manufacturing was still expanding in some U.S. regions.

Interpretation. This is useful falsification evidence against the idea that the measure is just proxying for long-run declining places.

4.6 Alternative channels considered by the paper

The paper explicitly considers several other explanations.

- Lower male presence could reflect migration, incarceration, homelessness, or mortality.
- The paper cannot measure all of these cleanly, but it shows that mortality rises and that prior work finds little robust out-migration from import-competing CZs.
- Male distress may therefore extend beyond the narrow set of deaths captured in the data.

Interpretation. The reduction in “marriageable men” likely reflects broader deterioration than labor earnings alone.

4.7 What the paper can and cannot distinguish

A key limitation is that the paper cannot fully separate:

- Becker’s relative-status channel, from
- Wilson’s absolute-status channel.

Because China shocks reduce young men’s outcomes both absolutely and relative to women, the two mechanisms move in the same direction in this setting.

Interpretation. The paper is strongest on the existence and importance of the labor-market-to-family link, and less definitive on which conceptual framework deserves all the credit.

4.8 Relation to the literature

The paper links several literatures.

- It extends the China-shock literature from employment and earnings to family structure.
- It provides rare quasi-experimental evidence on Wilson’s “when work disappears” thesis.

- It complements Becker-style relative-income studies by using gender-specific labor-demand shocks rather than raw gender wage differences.

5 Tables and figures

5.1 Table 1: manufacturing employment, relative employment, earnings, and idleness

Read:

- Panel A shows that the overall China shock reduces manufacturing employment for both young men and young women by about one percentage point.
- Panel B shows that beyond manufacturing, men lose more overall employment than women: the male-female employment gap shrinks by 0.65 points and the male-female NILF gap rises by 0.46 points.
- Panel C shows compression of the male-female earnings gap: \$672 at the 25th percentile, \$445 at the median, and \$847 at the 75th percentile.
- Panel D shows that the rise in male nonparticipation is almost entirely an idleness effect rather than a schooling effect.
- The male-industry versus female-industry rows show the crucial sign reversal: male-intensive shocks hurt men’s relative outcomes, while female-intensive shocks offset them.

Economic message. The paper first establishes the causal labor-market channel: China trade shocks reduce men’s relative economic standing, especially at the bottom and in inactivity rather than education.

5.2 Figure 1: distributional earnings effects

Read:

- Panel A shows larger earnings losses for men than for women at nearly every point of the earnings distribution above the bottom.
- Panel B rescales the compression in the male-female earnings gap by baseline male earnings and shows that the proportional effect is strongest at low earnings percentiles.

Economic message. The shock is especially damaging for lower-earning young men. This makes the decline in “marriage-market value” particularly plausible, because it hits the group already closest to economic insecurity.

5.3 Table 2: male availability and mortality

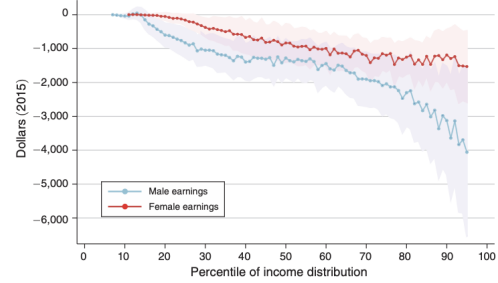
Read:

TABLE 1—ESTIMATED IMPACT OF MANUFACTURING TRADE SHOCK ON MANUFACTURING EMPLOYMENT BY GENDER AND GENDER DIFFERENTIAL IN EMPLOYMENT STATUS, EARNINGS, AND IDLENESS, 1990–2014: 2SLS ESTIMATES

	Panel A. Manufacturing employment as a share of population, age 18–39			Panel B. Male-female differential by employment status, age 18–39		
	M + F (1)	Males (2)	Females (3)	Emp (1)	Unemp (2)	NILF (3)
<i>I. Overall trade shock</i>						
Δ Import penetration	–1.06 (0.17)	–0.99 (0.17)	–1.09 (0.20)	–0.65 (0.26)	0.19 (0.09)	0.46 (0.24)
<i>II. Male industry versus female industry shock</i>						
Δ Import penetration × (male ind emp share)	–1.21 (0.44)	–2.59 (0.51)	0.20 (0.43)	–3.13 (0.78)	0.38 (0.26)	2.75 (0.62)
Δ Import penetration × (female ind emp share)	–0.88 (0.35)	0.82 (0.46)	–2.56 (0.38)	2.17 (0.65)	–0.02 (0.26)	–2.15 (0.64)
Mean outcome variable level in 1990	–2.61 12.98	–3.19 17.37	–2.06 8.68	–2.74 14.64	0.03 1.22	2.71 –15.87
	Panel C. Male-female differential in annual earnings (\$), age 18–39			Panel D. Male-female differential in idleness, age 18–25		
	P25 (1)	Median (2)	P75 (3)	Emp (1)	No emp in school (2)	No emp no school (3)
<i>I. Overall trade shock</i>						
Δ Import penetration	–672 (193)	–445 (191)	–847 (334)	–0.64 (0.34)	–0.02 (0.26)	0.66 (0.20)
<i>II. Male industry versus female industry shock</i>						
Δ Import penetration × (male ind emp share)	–2.216 (516)	–2.945 (593)	–3.685 (1,081)	–3.16 (1.03)	0.56 (0.73)	2.60 (0.60)
Δ Import penetration × (female ind emp share)	1.086 (529)	2.400 (630)	2.384 (814)	2.24 (0.92)	–0.68 (0.74)	–1.55 (0.56)
Mean outcome variable level in 1990	–1.894 6,926	–2.126 13,376	–2.491 17,489	–2.83 7.70	–0.25 0.87	3.08 –8.56

Notes: Observations = 1,444 (722 CZ × 2 time periods). Dependent variables: change in percentage of population age 18–39 that is employed in manufacturing; change in gender differentials in employment status (in percent pts); change in gender differential in annual earnings (in dollars); change in gender differential in percentage of young adults age 18–25 that are employed, not employed but in school, or neither employed nor in school. Panel C analyzes the change over time in the difference between a percentile of the unconditional male earnings distribution in a commuting zone and the corresponding percentile in the unconditional female earnings distribution. All models include a dummy for the 2000–2014 period, occupational composition controls (start-of-period indices of employment in routine occupations and of employment in offshorable occupations as defined in Autor and Dorn 2013), start-of-period shares of commuting zone population that is Hispanic, black, Asian, other race, foreign born, and college educated, as well as the fraction of women who are employed. Models are weighted by the product of period length and commuting zone share of start-of-period US mainland population. Robust standard errors in parentheses are clustered on state.

Panel A. Impact on male and female annual earnings by percentile, 1990–2014



Panel B. Impact on male-female annual earnings gap 1990–2014 as a percentage of 1990 male earnings

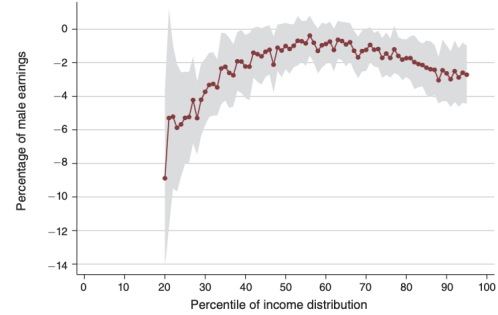


FIGURE 1. IMPACT OF MANUFACTURING TRADE SHOCK ON EARNINGS OF MALES AND FEMALES AGE 18–39, 1990–2014

Notes: Panel A measures the impact of a unit trade shock on the unconditional distribution of annual earnings (in \$2015) separately for males and females. Each dot indicates a coefficient estimate from a separate IV quantile regression with group-level treatment (Chetverikov, Larsen, and Palmer 2016) that controls for the covariates indicated in Table 1, and shaded areas indicate a 95 percent confidence interval. Panel B reports the effect of a unit trade shock on the difference in the male-female annual earnings gap expressed as a percentage of male earnings in 1990 at the indicated percentile.

- The male share of the young-adult population falls by about 0.25 percentage points after a one-unit trade shock.
- Excess male mortality rises by 64.4 per 100,000 per decade.
- The mortality increase is concentrated in drug and alcohol poisoning, HIV/AIDS, and homicide, not in suicide or accidents.
- Male-intensive shocks have much larger effects than female-intensive shocks.

Economic message. Trade shocks do not merely lower wages. They also reduce the supply and social functioning of young men in local marriage markets.

5.4 Table 3: marriage, fertility, mothers' status, child poverty, and household structure

Read:

- Marriage falls and never-marriage rises for women ages 18–39.
- Fertility falls, but less than marriage, so the share of mothers who are unmarried rises.
- Child poverty increases and the fraction of children in married-couple households falls.
- The adverse effects are driven by shocks to male employment, while female-employment shocks move outcomes in the opposite direction.

TABLE 2—ESTIMATED IMPACT OF MANUFACTURING TRADE SHOCK ON MALE SHARE OF ADULT RESIDENTS AND GENDER DIFFERENTIALS IN DEATH RATES 1990–2015: 2SLS ESTIMATES

	Panel A. Percentage of male residents		Panel B. Male-female death rate differential ages 20–39						
	18–39	18–25	Total	D&A Poison	HIV	Homicide	Suicide	Accident	All other
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>I. Overall trade shock</i>									
Δ Import penetration	−0.25 (0.11)	−0.28 (0.16)	64.4 (22.3)	19.5 (6.7)	21.6 (8.6)	14.0 (8.5)	−2.4 (4.3)	4.0 (8.4)	7.7 (5.5)
<i>II. Male versus female industry shock</i>									
Δ Import penetration	−0.62 (0.27)	−0.76 (0.43)	189.7 (60.0)	60.3 (23.6)	66.5 (20.3)	103.0 (27.5)	−9.8 (11.1)	−40.2 (27.2)	9.9 (16.3)
× (male ind emp share)									
Δ Import penetration	0.18 (0.15)	0.26 (0.29)	−77.1 (49.0)	−26.6 (18.9)	−29.1 (15.5)	−86.4 (29.2)	6.0 (10.9)	53.9 (27.1)	5.1 (16.7)
× (female ind emp share)									
<i>III. Summary stats: cumulative mortality 1990–2015 (decadal averages)</i>									
Male-female gap	936.0	93.8	110.3	154.4	168.9	262.5	146.10		
Males	1,644.6	153.3	146.6	198.6	218.6	378.6	548.92		
Females	708.7	59.5	36.3	44.3	49.7	116.1	402.82		

Notes: Observations = 1,444 (722 CZ × 2 time periods). Dependent variables: Change in percentage of male share of residents; cumulative male-female difference in death rates per 100,000 population by cause of death. The percentage of male residents is measured for the period 1990–2014 among all individuals who do not reside in institutionalized group quarters. Male share of CZ residents in 1990 was 49.6 percent among ages 18–39 and 50.2 percent among ages 18–25. Weighted mean changes in these variables were 0.11 and 0.19, respectively. Cumulative decadal mortality rates cover the period 1990–2015. All regressions include the full set of control variables from Table 1, and regressions in panel B control for a ten-year lag of the male-female differential in total mortality. Regressions are weighted by the product of period length and CZ population share, and standard errors are clustered on state.

TABLE 3—ESTIMATED IMPACT OF MANUFACTURING TRADE SHOCK ON MARRIAGE, FERTILITY, MATERNAL STATUS, CHILDHOOD POVERTY, AND HOUSEHOLD STRUCTURES OF ADULT WOMEN AND DEPENDENT CHILDREN, 1990–2014: 2SLS ESTIMATES

	Panel A. Women's marital status			Panel B. Fertility and maternity			Panel C. Percent of children in HH < poverty line
	Married	Widowed divorced separated	Never married	Births per 1,000 women	Percent of women w/ children	Percent mothers unmarried	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>I. Overall trade shock</i>							
Δ Import penetration	−0.95 (0.30)	−0.21 (0.11)	1.16 (0.33)	−1.54 (0.37)	−0.66 (0.23)	0.52 (0.31)	0.61 (0.26)
<i>II. Male versus female industry shock</i>							
Δ Import penetration	−3.57 (0.62)	−0.66 (0.22)	4.23 (0.64)	−4.65 (0.84)	−1.79 (0.63)	3.28 (0.73)	2.13 (0.70)
× (male share)							
Δ Import penetration	2.03 (0.55)	0.29 (0.19)	−2.32 (0.58)	2.01 (0.87)	0.62 (0.52)	−2.62 (0.85)	−1.12 (0.82)
× (female share)							
Mean outcome variable	−6.92	−1.62	8.55	−1.45	−3.53	6.56	1.65
Level in 1990	53.05	12.11	34.84	86.87	53.24	23.98	17.99
<i>Panel D. Women's household type</i>							
	Living w/ spouse	Living w/ partner	Other HH structure	Married couple	Parent + unmarried partner	Single parent, no partner	Grandparent or other
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>I. Overall trade shock</i>							
Δ Import penetration	−0.81 (0.27)	−0.22 (0.12)	1.03 (0.30)	−0.35 (0.19)	−0.11 (0.07)	0.30 (0.11)	0.15 (0.16)
<i>II. Male versus female industry shock</i>							
Δ Import penetration	−3.21 (0.55)	0.04 (0.28)	3.17 (0.60)	−1.85 (0.50)	0.28 (0.23)	1.43 (0.32)	0.14 (0.42)
× (male share)							
Δ Import penetration	1.93 (0.54)	−0.52 (0.20)	−1.41 (0.52)	1.36 (0.55)	−0.55 (0.25)	−0.98 (0.42)	0.17 (0.29)
× (female share)							
Mean outcome variable	−7.57	1.65	5.93	−4.69	1.62	1.79	1.28
Level in 1990	50.30	5.25	44.45	71.39	2.82	16.82	8.96
Poverty rate 1990	n/a	n/a	n/a	8.7%	42.3%	47.4%	28.8%

Notes: Observations = 1,444 (722 CZ × 2 time periods). Dependent variables: changes in women's marital status, births per 1,000 women, fraction of women with children, and fraction of mothers unmarried; fraction of children living in poverty; and household type of women and children. Outcomes in panels A, B, and D consider adult women ages 18–39 while those in panels C and E consider children ages 0–17. Fertility in panel B, column 4 is measured through 2016 while all other outcomes are measured through 2014. Dependent variables are: fraction of women with any biological, adopted, or stepchildren in the household (panel B, column 5); fraction currently married among women with children in the household (panel B, column 6); the fraction of children in households below the official census poverty line (panel C, column 7). Panel D, columns 1 and 2, refer to households where either (1) the woman is the spouse or partner of the household head or (2) she is the household head and has a spouse or partner who is living in the household. Panel D, column 3, comprises all other household structures. Dependent variables in panel E, columns 4–7 are the fraction of children in each household type: household head is a married parent of the child (panel E, column 4); household head is a parent with cohabiting partner (panel E, column 5); household head is a single parent (panel E, column 6); or household head is a grandparent, other relative, or non-related caregiver (panel E, column 7). All regressions include the full set of control variables from Table 1, and regressions are weighted by the product of period length and CZ population share, and standard errors are clustered on state.

Economic message. The family results are not generic recession effects. They line up closely with a mechanism in which diminished male earnings capacity reduces marriage and stable joint child-rearing.

5.5 Conclusion paragraph and implied magnitudes

Read:

- Using the observed rise in China import penetration from 1990 to 2014, the authors infer that trade pressure can explain about 3.9 percentage points of the observed 7.2-point decline in young male employment.
- The same calculation implies a 2.4-point reduction in marriage among young women, compared with an observed decline of 16.7 points.
- It also implies a 1.6-point increase in child poverty, compared with an observed increase of 4.0 points.

Economic message. Trade shocks are not the whole story, but they are quantitatively large enough to account for a meaningful share of the long-run deterioration in work and family outcomes.

6 Short summary

- The paper studies whether manufacturing decline lowers the marriage-market value of young men.
- It uses China-import exposure across commuting zones and decomposes that shock by male versus female industry intensity.
- Trade shocks reduce male relative employment and earnings, increase male idleness and premature mortality, lower marriage and fertility, and worsen children's household circumstances.
- The sign reversal between male-intensive and female-intensive shocks is central: shocks to men's jobs are socially damaging, shocks to women's jobs often have the opposite family effects.
- The paper cannot fully separate Becker's relative-income mechanism from Wilson's absolute-status mechanism, but it strongly supports the broader claim that when stable blue-collar work disappears, family structure changes.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that labor-demand shocks hitting male-intensive manufacturing reduce the economic standing of young men relative to women and that these changes spill over into marriage, fertility, and child poverty. The evidence is strongest because the same empirical design explains both labor-market and family outcomes, and because male- and female-intensive shocks move family outcomes in opposite directions.

7.2 Economic intuition

Marriage and joint child-rearing are more attractive when potential male partners have stable employment and earnings. Trade shocks that erode that stability do not just reduce men's wages; they raise idleness, lower male presence in local marriage markets, and increase visible distress. Those changes reduce marriageability and make single-parent family structures more common.

7.3 Takeaway

This paper is micro-to-macro demonstrations that labor-market structure matters for family structure. It shows that the social consequences of trade shocks run through gendered labor-demand channels, not merely through aggregate income loss.